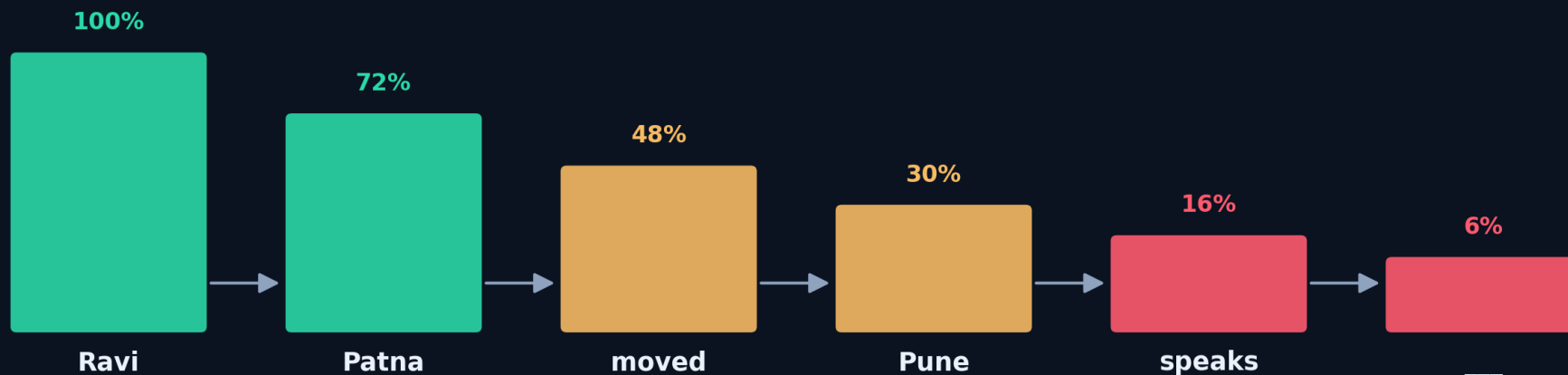


By word 50, word 1 is **almost gone**

MEMORY THAT LEAKS

"Ravi, who grew up in Patna and later moved to Pune, speaks ____"

how much of each word still survives in memory



By the blank, "Patna" is nearly gone — so it cannot tell you Hindi

That fading is the vanishing gradient. Attention was invented to fix it.

- 1 'Ravi, who grew up in Patna and later moved to Pune, speaks ____'
- 2 Each word updates a small memory passed to the next step.
- 3 By the blank, 'Patna' is down to 6%. It cannot tell you Hindi.

REMEMBER THIS

That fading is the vanishing gradient. Attention was invented to fix it.